Mikhail Kuznetsov

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Summary

Research Scientist specializing in foundation models for audit/security logs, interpretable/robust self-supervised learning, and AI security. Led the audit-log FM end-to-end: diversity-preserving 10B → 10M training-data sampling, pretraining + contrastive fine-tuning, interpretable anomaly scoring, and evaluation with limited feedback. Current work includes LLM alignment and securing agentic AI systems.

Research Interests

Agentic AI security • Foundation models for logs • Robust/self-supervised learning • Anomaly detection • Representation learning for structured/tabular data • Evaluation & interpretability • LLM alignment

Experience

Amazon AWS – AI Security and Observability *2022 – Present*
Senior Applied Scientist (promoted on accelerated timeline)

- **Agentic AI Security**: Leading efforts of securing AI agents, developing detection models on Agentic traces (e.g. Amazon Bedrock AgentCore). The models are focused on detecting prompt injection through unusual patterns of tool usage, given their historical observations and agentic context (user/system prompts).
- **LLM-Based Foundation Model on Audit Logs**: Tech lead of a foundation model for audit logs, learning embeddings for security entities (IP, API, usernames) from dynamic audit data. The model was deployed to production in Sep 2025, enhancing Customer-facing anomaly detection, reducing False Positives by 20–30%. Main contributions are:

  Training data creation. Designed a sampling algorithm that compresses ∼10B raw log events to ∼10M high-value examples while preserving entity diversity; runs on a regular cadence to continually refresh training data.

  Contrastive Fine-tuning. Introduced contrastive fine-tuning for tenant-specific contexts, which are focused on entities particularly important for downstream consumers (asnOrg, api).

  Evaluation & interpretability. Built a multi-signal evaluation framework (weak supervision, third-party feeds, expert labels) and interpretability tooling (embedding similarity, contextual neighbors) for explainable anomalies.

  Architectural design (hierarchical logs). Co-designed a memory-efficient architecture leveraging the nested structure of log entries for long-context, high-throughput security workload.

  GenAI in Security Benchmarking: Co-authored a benchmark evaluating generative models in security workflows.
- **Robust/Secure Learning**: Co-developed distributionally robust self-supervised learning for tabular data; advanced graph-based domain reputation with weak supervision; ongoing research on agentic AI security for context-aware anomaly/failure detection in multi-agent systems.

Yahoo! Research, New York *2017 – 2021*
Research Scientist

- **ExtremeText**: contributed to a no-regret generalization of hierarchical softmax for *extreme* multi-label classification; improved training/inference efficiency and accuracy over very large label spaces.
- **VisualTextRank**: designed an unsupervised, graph-based content extraction method to automate ad text → image matching; formulated a TextRank-style ranking over multimodal (text/vision) features, built data/eval pipelines, and improved retrieval relevance in internal benchmarks.

- Built unified multi-task CTR/CVR models that improved advertising ROI by >10% in online A/B tests.

Education

Moscow Institute of Physics and Technology
Ph.D. (Computer Science, Mathematical Modeling), 2016
M.S. with Honors, Applied Mathematics & Physics, 2013
B.S. with Honors, Applied Mathematics & Physics, 2011

Yandex School of Data Analysis — Diploma in Data Science, 2012

Publications by Topic

Foundation Models for Logs & Security

- *HLogformer: A Hierarchical Transformer for Representing Log Data.* Z. Hou, M. Ghashami, **M. Kuznetsov**, MA. Torkamani. 2024
- *An Evaluation Benchmark for Generative AI in Security Domain.* M. Ghashami, **M. Kuznetsov**, V. Gao, G. Teng, P. Wallis, J. Xie, A. Torkamani, B. Coskun, W. Ding. KDD GenAI Evaluation Workshop, 2024.

Robust / Extreme Classification

- *Distributionally Robust Self-Supervised Learning for Tabular Data.* S. Ghosh, J. Xie, **M. Kuznetsov**. NeurIPS Third Table Representation Learning Workshop, 2024.
- *A No-Regret Generalization of Hierarchical Softmax to Extreme Multi-Label Classification.* M. Wydmuch, K. Jasinska, **M. Kuznetsov**, R. Busa-Fekete, K. Dembczynski. NeurIPS, 2018.

Language Model Alignment

- *STARS: Synchronous Token Alignment for Robust Supervision in Large Language Models.* M. Quamar, M. Areeb, **M. Kuznetsov**, M. Ozmen, B. Celik. SPIGM Workshop, ICML, 2026.
- *Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models.* M. Quamar, M. Areeb, N. Sharma, Nishant, A. Shreekumar, J. Jonathan, M. Ozmen, **M. Kuznetsov**, B. Celik. 2025.

Generative Models & Retrieval for Ads/Vision

- *Salient Object-Aware Background Generation using Text-Guided Diffusion Models.* A. Eshratifar, J. Soares, K. Thadani, S. Mishra, **M. Kuznetsov**, Y. Ku, P. De Juan. CVPR Workshop on Generative Models for Computer Vision, 2024.
- *Staging E-Commerce Products for Online Advertising using Retrieval Assisted Image Generation.* Y. N. Ku, **M. Kuznetsov**, S. Mishra, P. de Juan. AdKDD, 2023.
- *VisualTextRank: Unsupervised Graph-Based Content Extraction for Automating Ad Text-to-Image Search.* KDD (ADS Track). S. Mishra, **M. Kuznetsov**, G. Srivastava, M. Sviridenko, 2021.

Skills

Data & Infra: PySpark; Pandas; Arrow/Parquet; large-scale data curation/sampling; AWS (SageMaker, S3, EMR, EKS).

Modeling: scikit-learn; PyTorch; Hugging Face (Transformers, Accelerate, Datasets, PEFT); vLLM; LoRA/QLoRA.